

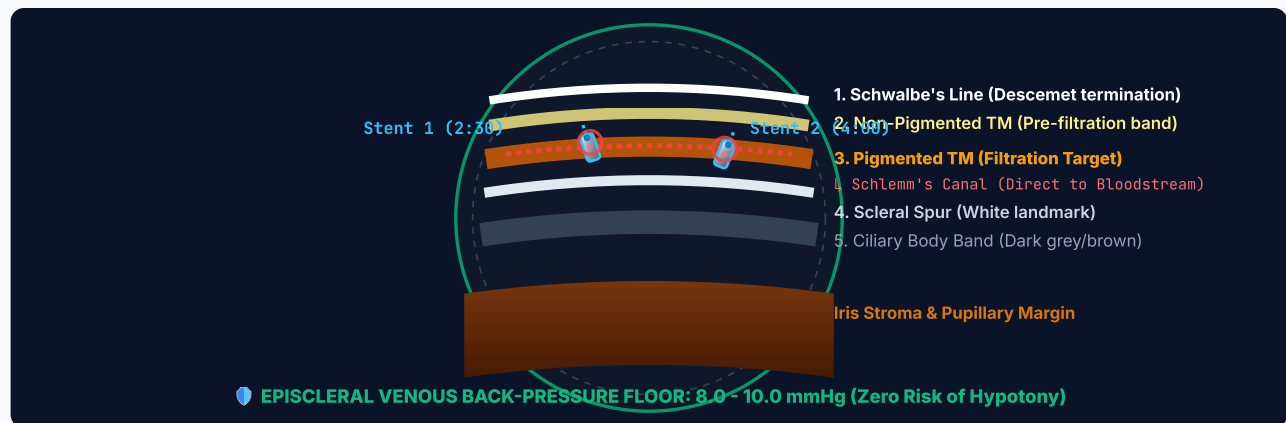
MIGS: Trabecular Micro-Bypass Stent Surgery

Direct-to-Bloodstream Canalicular Drainage & Physiological 8–10 mmHg Back-Pressure Floor

☀ What Is a Trabecular Micro-Bypass Stent in Simple Terms?

In open-angle glaucoma, the eye's natural drainage filter—called the **trabecular meshwork**—becomes clogged with microscopic debris, raising intraocular pressure (IOP) to 30+ mmHg and crushing the optic nerve fibers. Instead of cutting open the outside of the eye (like risky older surgeries), we insert microscopic titanium snouts (stents) through the clogged filter directly into **Schlemm's canal** and collector veins. Because Schlemm's canal empties directly into the venous bloodstream, the venous blood establishes a rigid **8 to 10 mmHg backpressure floor**. The eye's pressure can never drop below 8 mmHg, preventing flat anterior chambers and dangerous low pressure (hypotony)!

DIRECT GONIOSCOPIC ANATOMY, TRABECULAR LANDMARKS & DUAL TITANIUM STENT PLACEMENTS



STEP 1 Microscope (38°) & Patient Head Tilt (35°) — Overcoming Total Internal Reflection

Tool: Zeiss OPMI Lumera
Goniometric Arm

👉 **What To Do Right Now:** Tilt the operating microscope 35° to 40° toward yourself, and rotate the patient's head 30° to 35° away from you. This creates a combined ~70° optical angle of incidence.

💡 **Why This Step Is Strictly Necessary:** Light rays coming from the drainage angle hit the curved cornea at an angle greater than 46° (the critical angle), causing *total internal reflection*. Without tilting both the microscope and the patient's head, the drainage angle is 100% invisible—the cornea reflects light like a mirror!

⚠ **Critical Hazard:** Attempting to insert instruments blindly without goniometric tilt leads to iris shredding or lens capsule rupture.

**STEP
2****Direct Surgical Gonioprism Placement — Creating the Optical Viewport****Tool:** Swan-Jacob Direct Gonioprism

👉 **What To Do Right Now:** Apply a generous drop of viscoelastic onto the concave contact surface of the Swan-Jacob prism. Place it gently onto the temporal cornea without pressing down hard.

💡 **Why This Step Is Strictly Necessary:** The gonioprism neutralizes corneal curvature. Viscoelastic serves as an optical coupling agent, eliminating air bubbles and allowing light to pass directly into the surgeon's eyepieces.

⚠️ **Critical Hazard:** Pushing too hard on the cornea creates corneal folds (Descemet wrinkles) that distort angle visualization.

**STEP
3****Cohesive OVD Angle Deepening — Opening the Drainage Runway****Tool:** Provisc 10mg/mL Cohesive Cannula

👉 **What To Do Right Now:** Introduce the blunt cannula across the anterior chamber to the nasal angle. Inject 0.3 mL of cohesive jelly directly over the iris root to gently push the iris backward.

💡 **Why This Step Is Strictly Necessary:** In eyes with high pressure, the peripheral iris bows forward, crowding the trabecular meshwork. Viscoelastic creates a 1.5mm wide open canyon, exposing the golden-brown filtration band so the stent injector has clear passage without snagging iris tissue.

⚠️ **Critical Hazard:** Failing to deepen the angle risks snagging iris blood vessels, causing a massive hyphema (bleeding).

**STEP
4****Micro-Stent 1 Insertion (2:30 o'clock) — Direct Bypass into Schlemm's Canal****Tool:** Multi-Dose Stent Injector (iStent inject W)

👉 **What To Do Right Now:** Under high gonioscopic magnification, advance the injector tip across the pupil to the nasal angle at the 2:30 o'clock position. Approach the pigmented trabecular meshwork at a 15° to 20° angle. Gently press the trocar until the sleeve touches the TM, then depress the delivery button to deploy the first titanium stent.

💡 **Why This Step Is Strictly Necessary:** The stent's 360-micron pointed head penetrates the resistance-heavy trabecular meshwork, seating its intake lumen directly inside Schlemm's canal while its wide flange rests safely in the anterior chamber. Fluid can now bypass the blocked filter completely.

⚠️ **Critical Hazard:** Entering too posterior penetrates the scleral spur or ciliary body, causing severe pain and suprachoroidal hemorrhage.

**STEP
5****Micro-Stent 2 Insertion (4:00 o'clock) — Dual Stent Circumferential Outflow****Tool:** Multi-Dose Stent Injector (Second Stent)

👉 **What To Do Right Now:** Retract the inserter tip slightly, pivot the trocar 2 to 3 clock hours inferiorly to the 4:00 o'clock position, and deploy the second micro-stent into the pigmented TM.

💡 **Why This Step Is Strictly Necessary:** A single stent accesses only ~60° of Schlemm's canal. Deploying two stents separated by 2 to 3 clock hours recruits collector channels across the entire inferior-nasal quadrant, doubling outflow facility from 0.08 to >0.26 $\mu\text{L}/\text{min}/\text{mmHg}$ and ensuring durable pressure reduction.

⚠️ **Critical Hazard:** Placing stents too close together (<1 clock hour) taps the same collector channel and provides zero additional pressure drop.

👉 **What To Do Right Now:** Briefly decompress the anterior chamber by depressing the posterior lip of the wound or activating foot pedal position 2 (aspiration). Watch closely through the gonioscope: a crimson plume of venous blood will flow backward out of the stent lumen into the eye. Then, perform gentle irrigation/aspiration to wash out all viscoelastic.

💡 **Why This Step Is Strictly Necessary:** The blood reflux wave is the gold standard clinical proof that the stent is patent and connected directly to the episcleral venous bloodstream! When AC pressure is lowered below 8.5 mmHg, blood naturally rushes backward. Thorough viscoelastic washout prevents acute postoperative IOP spikes (>40 mmHg).

The Goldmann Equation & The 8–10 mmHg Back-Pressure Floor:

$$IOP = (F - U) / C + EVP$$

- **F** (Aqueous production) = 2.4 $\mu\text{L}/\text{min}$ | **U** (Uveoscleral outflow) = 0.35 $\mu\text{L}/\text{min}$
- **C** (Outflow facility) increases from **0.075** (diseased) to **0.28 $\mu\text{L}/\text{min}/\text{mmHg}$** (post-stent)
- **EVP (Episcleral Venous Pressure) = 8.5 mmHg** (Physiological bloodstream floor)

Because the stents drain directly into veins, IOP can NEVER drop below EVP (8.5 mmHg), entirely eliminating the risk of hypotony!